Candidate Number……....

THE PRESIDENT'S OFFICE – RALG

TAHOSSA ROMBO DISTRICT EXAMINATION SYNDICATE

FORM FOUR MOCK EXAMINATION

032/2B **CHEMISTRY 2B (ACTUAL PARCTICAL)**

TIME: 2:30 Hours **August, 2024**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **twenty five (25)** marks
3. Write your **Examination Number** on every page of your answer sheet.
4. The following constants may be used:
   Atomic masses: H=1, C=12, O=16, Na=23, Cl=35.5
   GMV at S.T.P = $22.4dm^3$
   1Litre=$1dm^3$=$1000cm^3$

FOR ASSESSOR'S ONLY		
QUESTION NUMBER	**SCORE**	**MARKER'S INITIALS**
1		
2		
TOTAL		
CHEKER'S INITIALS		

1. You are provided with the following:
   MM: A solution made by dissolving $0.56dm^3$ of hydrogen chloride in distilled water at S.T.P to make $250cm^3$ of solution.
   NN: A solution of 2.86g of hydrated sodium carbonate ($Na_2CO_3.WH_2O$) in $0.25dm^3$ solution.
   MO: Methyl Orange indicator

Procedure

Put solution **MM** into the burette. Pipette $20cm^3$ *or* $25cm^3$ of solution **NN** into the titration flask and then add two drops of **MO** indicator into it. Titrate solution **MM** against **NN** until the end point is reached. Record the burette reading. Repeat the procedure to obtain three more readings and record in a tabular form.

Questions

a) …..$cm^3$of solution **NN** required …..$cm^3$of solution **MM** for complete neutralization.
b) What will happen if the burette and pipette are not rinsed with solutions?
c) Write a balanced chemical equation for the reaction between solutions **MM** and **NN.**
d) Workout on;

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i. Molarity of solution **MM**
ii. Molar concentration of solution **NN**
iii. Mass concentration of solution **NN**
iv. Number of molecules of water of crystallization in hydrated sodium carbonate ($Na_2CO_3.WH_2O$)

2. Sample **Y** contains one cation and one anion. You are required to carry out the guided systematic procedure in Table 1 in order to identify the cation and anion present in sample **Y** provided.

Table 1

S/N	Experiment	Observation	Inference
(a)	Observe sample **Y.**		
(b)	Heat small amount of solid sample **Y** in a dry test tube.		
(c)	Add dilute HCl in a test tube containing small amount of solid sample **Y.**		
(d)	Prepare a stock solution of solid sample **Y**. Divide the resulting solution into four portions, then add; (i) NaOH solution to the first portion drop-wise till in excess		
	(ii) Ammonia solution to the second portion drop-wise till in excess		
	(iii) KI solution to the third portion, warm then cool the mixture.		
	(iv) $FeSO_4$solution to the fourth portion followed by concentrated H_2SO_4 alongside the test tube.		

Conclusion

i. Cation present in sample **Y** is ……. While anion is ……
ii. The IUPAC name of sample **Y** is …….
iii. Write the balanced chemical equation that took place in experiments **(b)** and **(d) -iii.**